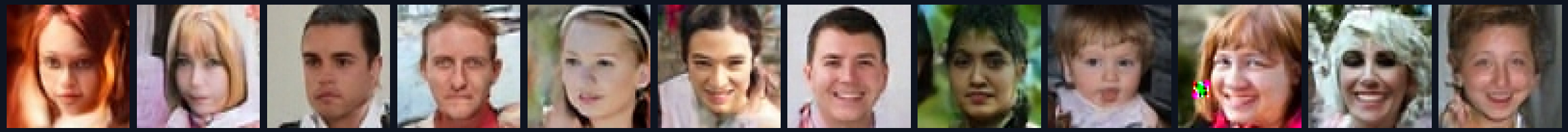
Frequency-AR image generation: what have we learned?



A1 (ours, 11.1M) · 250k EMA · UniPC [20,10,8,8] · std seeds 0–11, uncurated — seed 9 shows the rare single-token G3 checkerboard (slide 10)

THE MODEL

$$p(X) = \prod_k p(G_k \mid G_{<k})$$



G0	DC	UniPC 20 calls
G1	(0,1) (1,0)	UniPC 10 calls
G2	(2,0) (1,1) (0,2)	UniPC 8 calls
G3	(0,3) (1,2) (2,1)	UniPC 8 calls

Outer: AR over 4 groups of 4×4 block-DCT coefficients. Inner: a conditional diffusion ODE samples each group's whole 8×8 field (one axial transformer, exact prefix cache).

A1 freq-AR

9.46

FID-50K

params 11.1M
recall 0.413
46 serial calls
train 5.77 H100-h

R0 DCTdiff

8.68

FID-50K

params 44.2M
recall 0.440
15 serial calls
train 3.34 H100-h

Both: validated UniPC samplers, same q90 evaluator

NOT YET WON

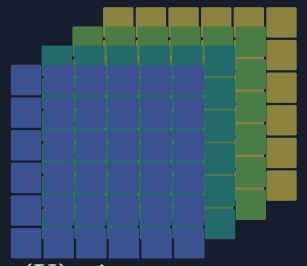
- **46 vs 15** serial evaluations
- **1.84×** latency, 1.36× GPU-s (H100)
- **1.72×** training H100-h for this FID
- **Recall gap** grows G1→G3

Open question Is explicit frequency causality a distinct, useful inductive bias — and can its per-field generator be made cheap enough to win the quality–compute frontier, not just the parameter count?

FFHQ64-q90 (70k), EDM Inception FID-50K + k=3 P/R; A1 a1_d256 250k EMA, R0 our DCTdiff port 250k EMA; latency: isolated H100 NVL bench (bs 500 / bs 16). efficiency/phase2_verdict.json, bench*.json

Why order image generation by frequency?

Joint diffusion



noise level t
shared by all

$p(X)$ via one reverse-time ODE over all of X

every position and every frequency is denoised together; coarse-to-fine emerges only implicitly

Spatial autoregression

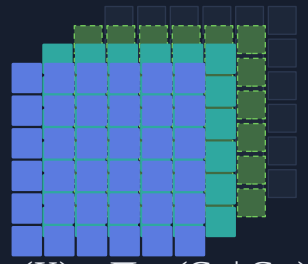


raster order

$p(X) = \prod_i p(x_i \mid x_{<i})$

pixels / patches in raster order; long serial chains, global layout decided late

Frequency autoregression (ours)

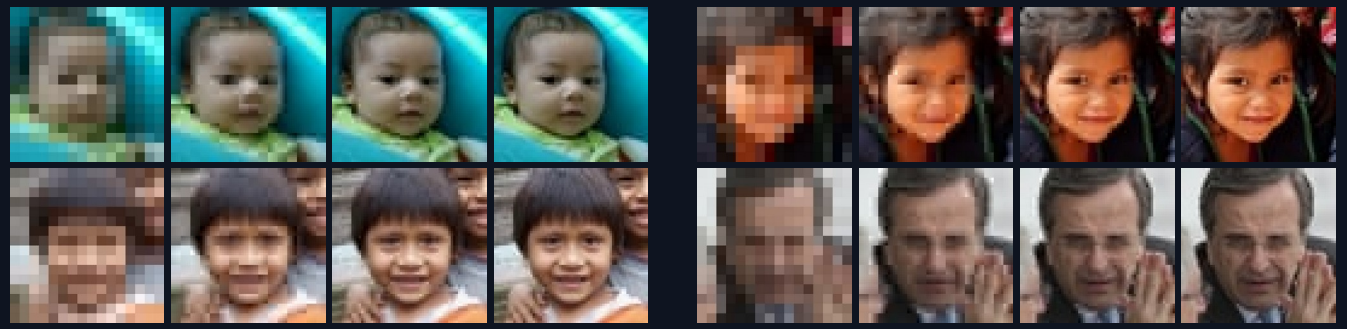


G0, G1 fixed
G2 being sampled
G3 next

$p(X) = \prod_k p(G_k \mid G_{<k})$

a whole spatial field per step, low → high frequency; each field sampled stochastically

Real FFHQ-64 images rebuilt by cumulative frequency groups (data, not model)



Real images: FFHQ64-q90 cache (real10k.pt, dataset index 7i), decoded with DCTdiff DCT_to_RGB after zeroing later groups. Group = u+v of the 4×4 zig-zag coefficient.

What the factorization offers

- A different causal order: layout (G0) is decided first, detail conditioned on it
- Explicit, inspectable separation of global vs fine information
- Questions, not assumptions: does it improve quality, conditional diversity, training or sampling cost?

No computational advantage is assumed; it is measured (slides 9, 12).

A1: one axial transformer, causal across frequency

1 · REPRESENTATION



x ∈ ℝ^(64×64×3)

YCbCr 4:2:0 → 4×4 block DCT → keep 9 zig-zag coefficients



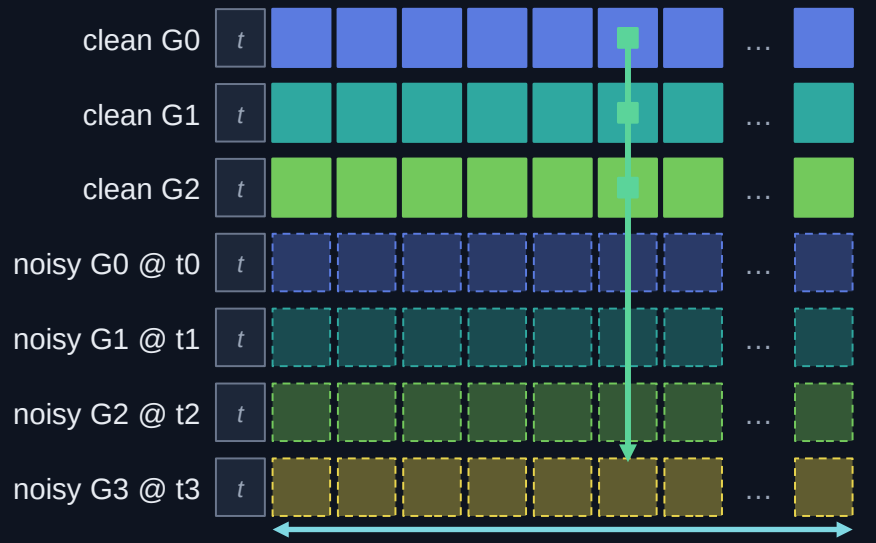
group = u + v
(3,0) and u+v ≥ 4 dropped

token = 8×8 px = 4 Y + Cb + Cr blocks × 9 coef = 54 ch; 64 tokens on an 8×8 grid

G0	6 ch / token · 384 values / image
G1	12 ch / token · 768 values / image
G2	18 ch / token · 1152 values / image
G3	18 ch / token · 1152 values / image

field $G_k \in \mathbb{R}^{(64 \times C_k)}$: 3456 values / image

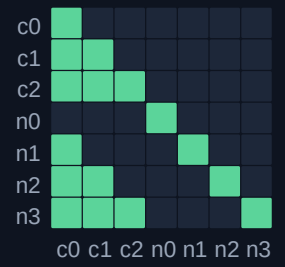
2 · TRUNK (TRAINING, ONE PASS)



spatial heads (3/4): dense, bidirectional over the 64 + 1 tokens of one slot

frequency head (1/4): same position across slots, masked →

frequency mask (row → column)



13 axial blocks (6·mid·6, long skips), D 256, 4×64 heads
Shared: all blocks, position + time embeddings
Per stage: adapters $\mathbb{R}^{C_k} \leftrightarrow \mathbb{R}^{256}$, stage embedding
[B, 7, 65, 256] · 11.09M params · ε-MSE

3 · SAMPLING (OUTER × INNER)

G0 — **G0 ~ p(G0)**
INNER: UniPC ODE, 20 calls

↓

G1 — **G1 ~ p(G1 | G<1)**
INNER: UniPC ODE, 10 calls · prefix K/V cached once

↓

G2 — **G2 ~ p(G2 | G<2)**
INNER: UniPC ODE, 8 calls · prefix K/V cached once

↓

G3 — **G3 ~ p(G3 | G<3)**
INNER: UniPC ODE, 8 calls · prefix K/V cached once

reassemble 54 ch → inverse 4×4 DCT → RGB

Inner diffusion acts only on the current field (64 × C_k); the outer chain never revisits earlier groups.

Masks/shapes verified: verify_axial*.json (clean G_k → $\hat{\varepsilon}(G_k)$ Jacobian exactly 0; cached vs training $\hat{\varepsilon} \le$ 1e-6). Training: 3 clean + 4 noisy slots, one t per stage. Sampler: efficiency/phase2_verdict.json.

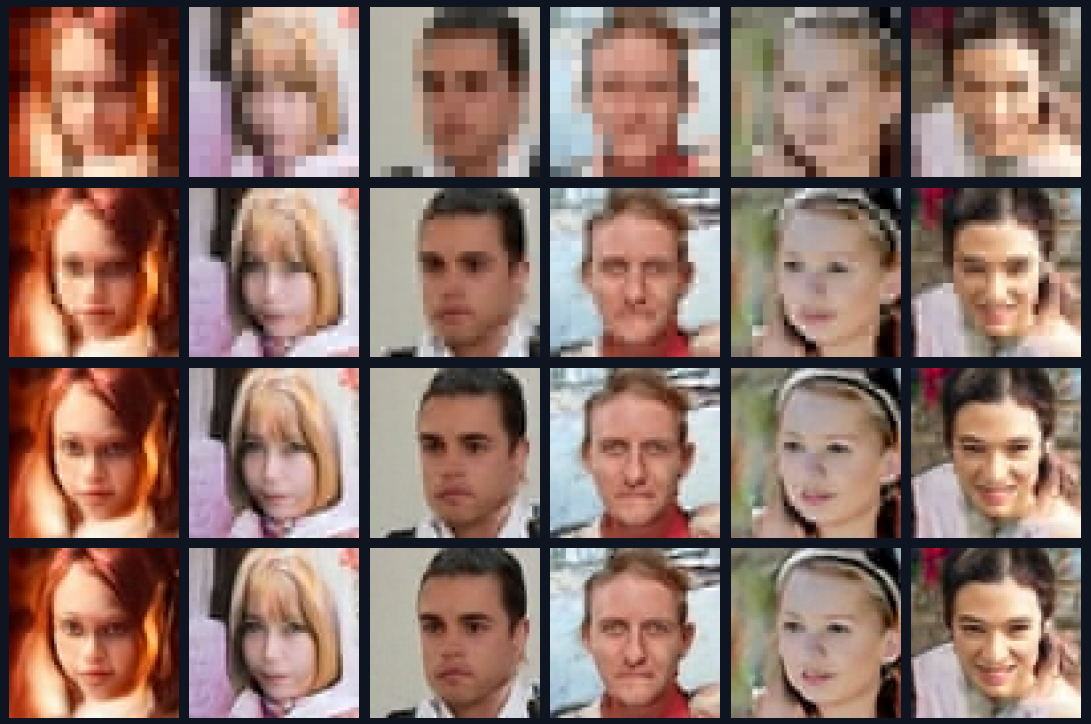
Next-frequency generation, stage by stage (real samples)

G0
`(0,0) DC`
+6 ch · 384 values

≤ G1
`+ (0,1) (1,0)`
+12 ch · 768

≤ G2
`+ (2,0) (1,1) (0,2)`
+18 ch · 1152

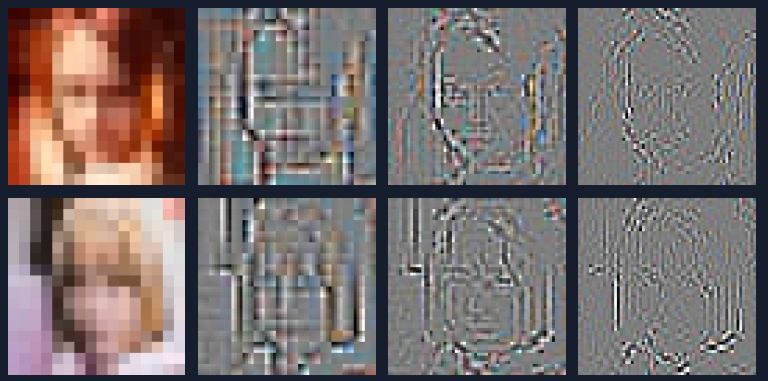
all
`+ (0,3) (1,2) (2,1)`
+18 ch · 1152



A1 250k EMA, UniPC [20,10,8,8]; each row decodes the generated coefficients of the groups so far; std seeds 0–5, uncurated

What each group alone contributes

band-only decodes (other groups zeroed), seeds 0–1



G0 G1 ×1.5 G2 ×2.8 G3 ×5

Re-sampling one group's noise (others fixed): share of image variation it controls

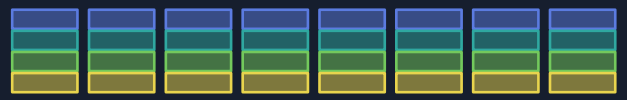
G0		pixel 98.2% · Inception 99%
G1		pixel 2.8% · Inception 31%
G2		pixel 0.8% · Inception 19%
G3		pixel 0.4% · Inception 12%

Samples: H100 run j20_10_8_8 (s0_coef.pt), decoded with DCT_to_RGB. Noise shares: efficiency/phase1_verdict.json (A1 250k, Euler, resample one stage's noise; Inception = mean sq. feature change vs independent draw).

Three references isolate what frequency causality does

R0 · DCTdiff (our port)

Joint denoising of all retained DCT coefficients

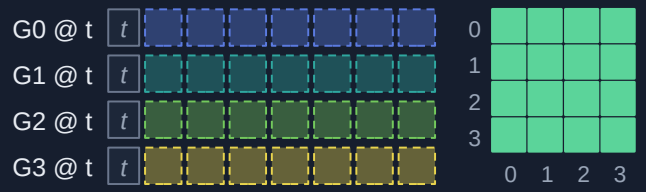


each of 64 tokens carries all 54 ch, noisy at one t

- U-ViT, D 512, 13 blocks (official FFHQ “small” recipe)
- one joint ODE: UniPC-15 or Euler-100
- 17.6 GFLOP / training image

A0 · joint axial

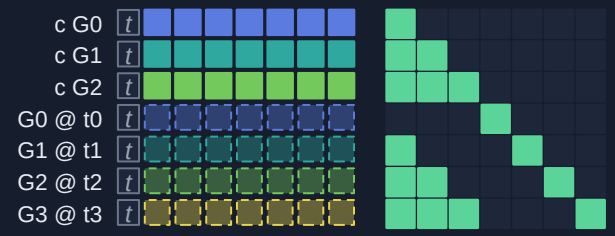
Our axial network, all groups noisy at one shared t



- bidirectional frequency attention
- one joint ODE (UniPC-15 best)
- 17.8 GFLOP / training image

A1 · frequency-AR axial

Same network, clean prefix + causal frequency mask



- causal frequency mask, t per stage
- 4 conditional ODEs (UniPC 46 calls)
- 31.1 GFLOP / training image

A1 − A0 the matched architectural test

identical blocks and parameter count; only slot set, mask and per-stage objective differ → effect of causal frequency factorization

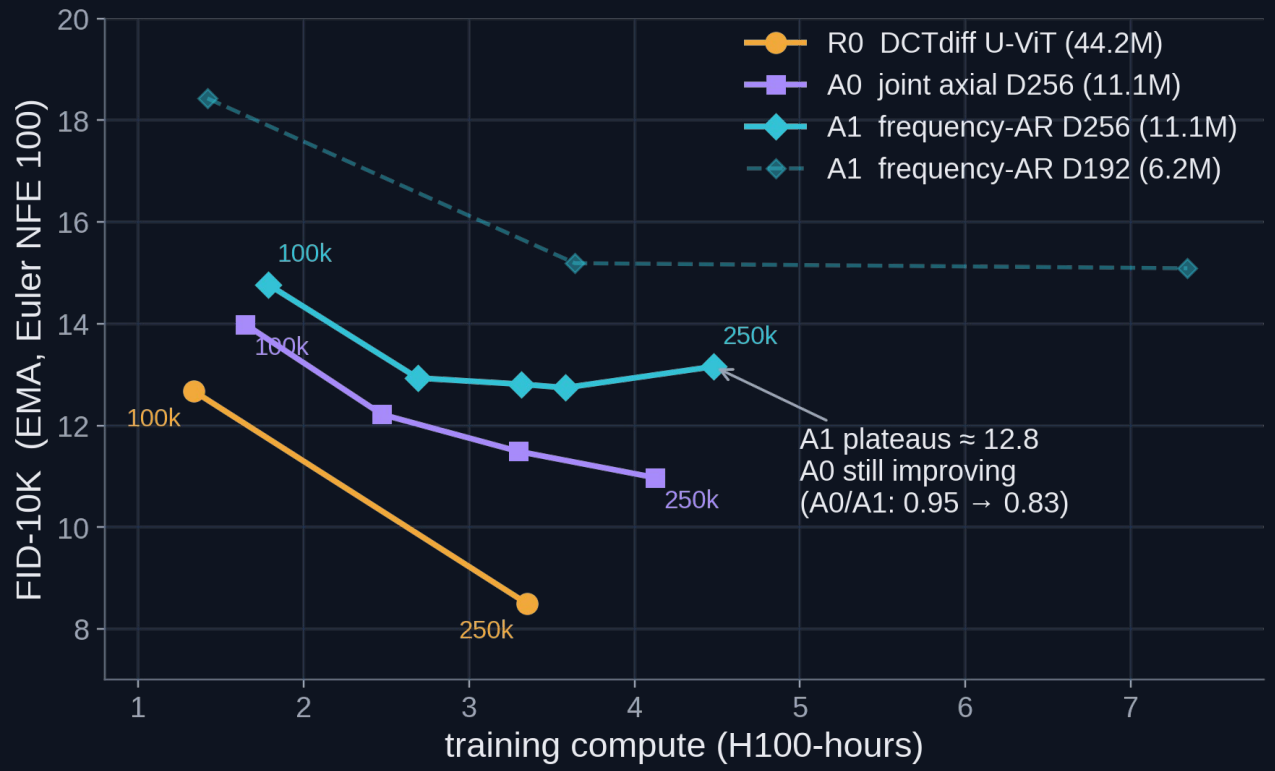
A1 − R0 practical competitiveness

confounded by 4× parameters, a different backbone and token design → never read as an architectural effect

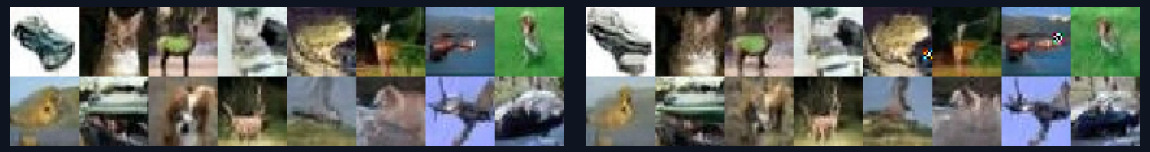
Earlier spatial-causal, 256-token staged and channel-mask designs were dropped (appendix A5).

Configs: third_party/DCTdiff/configs/ffhq64_exp.py (r0, a0_d256, a1_d256). Training FLOPs: FlopCounterMode, ffhq64/mature/summary.json.

CIFAR-10: causality looked nearly free, then plateaued



same seeds · official released DCTdiff 500k (left) vs A1 D192 500k (right): note single-token checkerboards

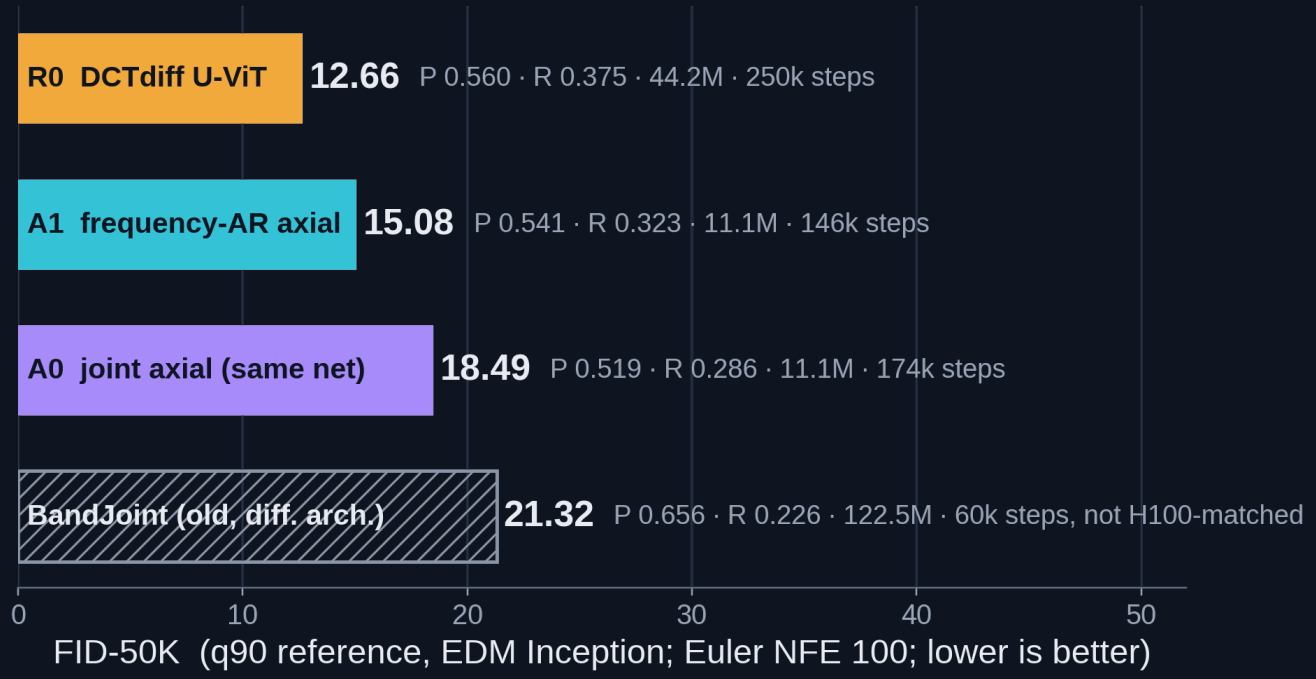


- **Early: A1 ≈ A0**
  A1/A0 = 1.05 at 100k (D192), recall −0.023
- **Mature: A0 kept improving**
  D256: A0/A1 0.95 → 0.83 at 250k; A1 plateau ≈ 12.8
- **Width helped**
  D192 → D256: 18.43 → 14.77 at 100k
- **More stages ≠ better**
  4×4 DCT K3 → K7: 0.95× FID at 2.06× H100-h
- **Full-image DCT overfit**
  early global bands memorise: FID 51.6, recall 0.25
- **G_{-1} context, stage noise**
  neutral (1.07×, 1.02× FID)

Takeaway on CIFAR, strict low→high ordering costs a growing share of quality at maturity, and the joint axial backbone is itself 1.35× R0 at equal H100-h. **Does this generalize to aligned faces?**

CIFAR-10 (DCTdiff's JPEG-q75 statistics), FID-10K, EMA, Euler NFE 100 per stage. scaling/summary.json (d256_maturity, arms, verdicts). EMA before 100k mostly measures lag, so curves start at 100k.

FFHQ-64: frequency AR beats its joint control





same std seeds 0–3 for every arm, Euler, uncurated

A1 > A0 on every matched axis but FLOPs

steps (FID-10K 20.09 vs 25.17 @100k) · H100-h (15.08 vs 18.49) · recall (.323 vs .286); per training FLOP ≈ tie (31.1 vs 17.8 GFLOP/img)

The CIFAR ranking reversed

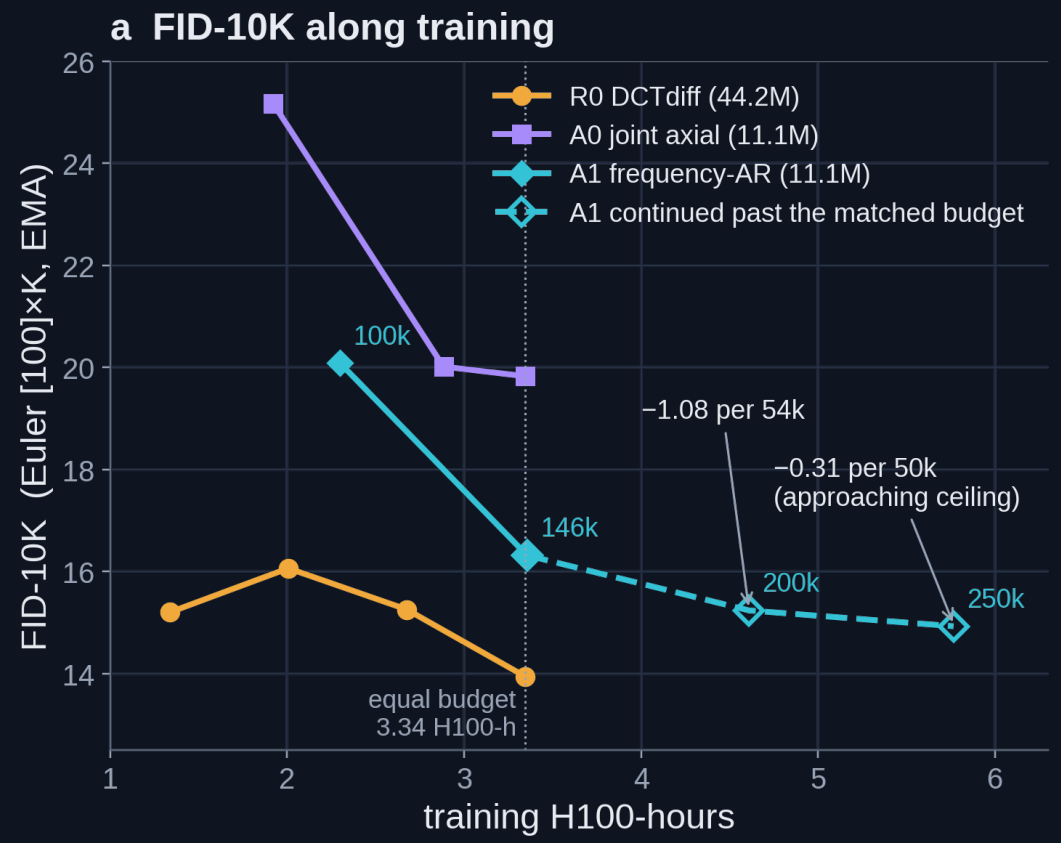
CIFAR: A0/A1 = 0.83 at 250k. FFHQ: A1/A0 = 0.82 at equal hours. Aligned faces may favour layout-first generation — one dataset, one seed per arm.

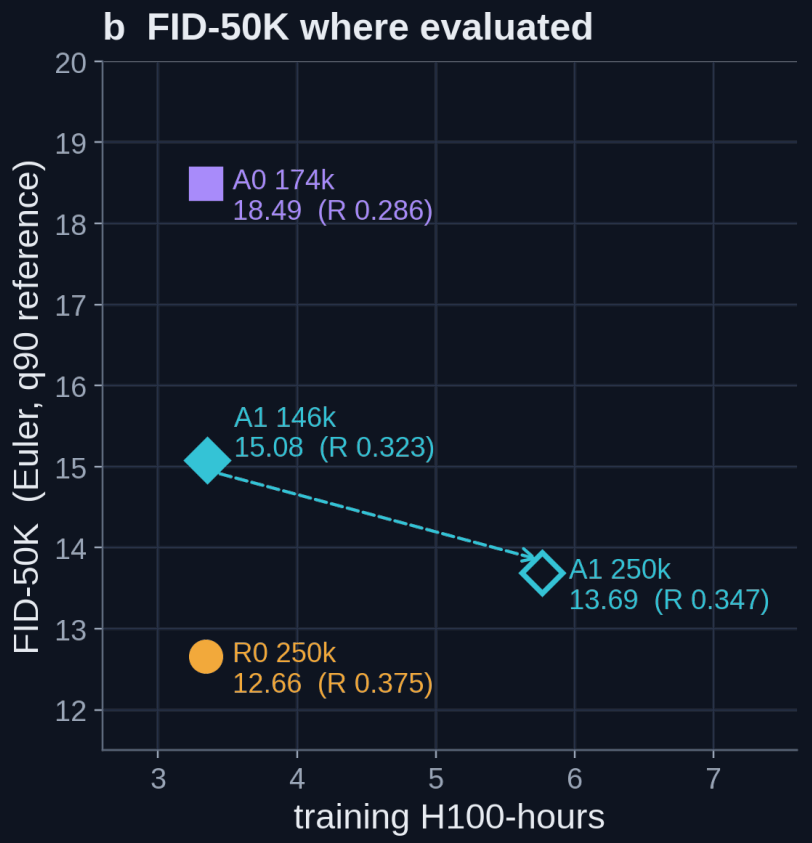
Not yet competitive with R0

A1 = 1.19× R0 FID-50K (pre-registered target ≤ 1.10×), still improving fast at 146k; sampling 1.80× R0's GPU-s at Euler.

FFHQ64-q90 (jpeg-ar q90, 70k), EDM Inception FID-50K, P/R k=3 at n=50k; EMA; Euler [100] per stage (A1 400 calls, R0/A0 100). ffhq64/summary.json · BandJoint winner: jpeg-ar campaign, same evaluator, 122.5M, its own sampler.

More training brought A1 close to R0, at higher cost





Undertrained at 146k

pre-registered 200k gate passed: FID-10K 16.32 → 15.24, recall .367 → .385, memorisation 0.03%

Approaching a ceiling

200k → 250k: −0.31 FID-10K for 1.16 H100-h; FID-50K 13.69 = 1.08× R0 (inside the 1.10× band)

No further extension

5.77 H100-h = 1.72× R0, recall .347 vs .375, G3 tail rate 2.9 → 3.6%: more exposure was not the lever

Euler [100]×K, EMA, q90 ref. a: FID-10K (A1 146k/200k on std seeds; R0/A0 finals = first 10k of their 50k runs). b: FID-50K only where run. ffhq64/summary.json, ffhq64/mature/summary.json, efficiency/baseline.json.

UniPC: 400 → 46 evaluations, FID-50K 13.75 → 9.46

Euler	`[100, 100, 100, 100] → 400`
UniPC	**`[ 20,  10,  8,  8] → 46`**

calls for G0, G1, G2, G3 · UniPC-2 bh2, logSNR grid, terminal Euler step

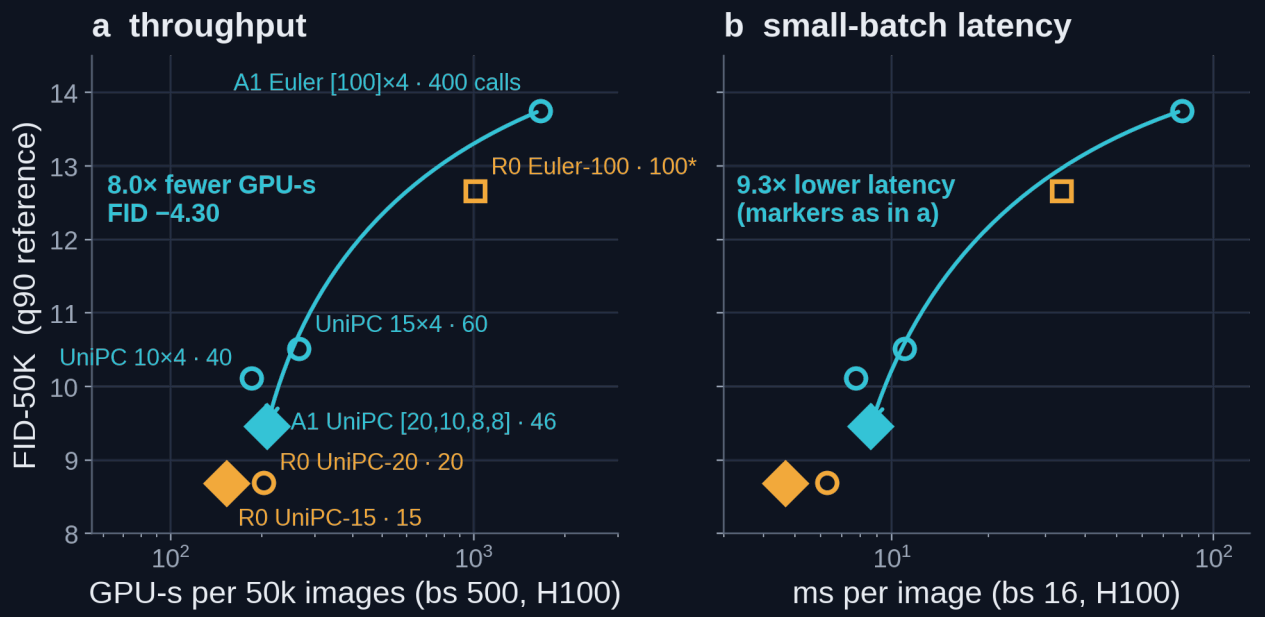
FID-50K
9.46
from 13.75 · −31%

recall
0.413
from 0.349 · +0.064

GPU-s / 50k
208
from 1662 · 8.0×

ms / image
8.6
from 79.9 · 9.3×

Allocation matters: G0 needs ≥ 15–20 calls (8 → FID-10K 13.8), G2/G3 tolerate 8; G0 errors amplify 3–4× downstream. Cuts compound non-additively.

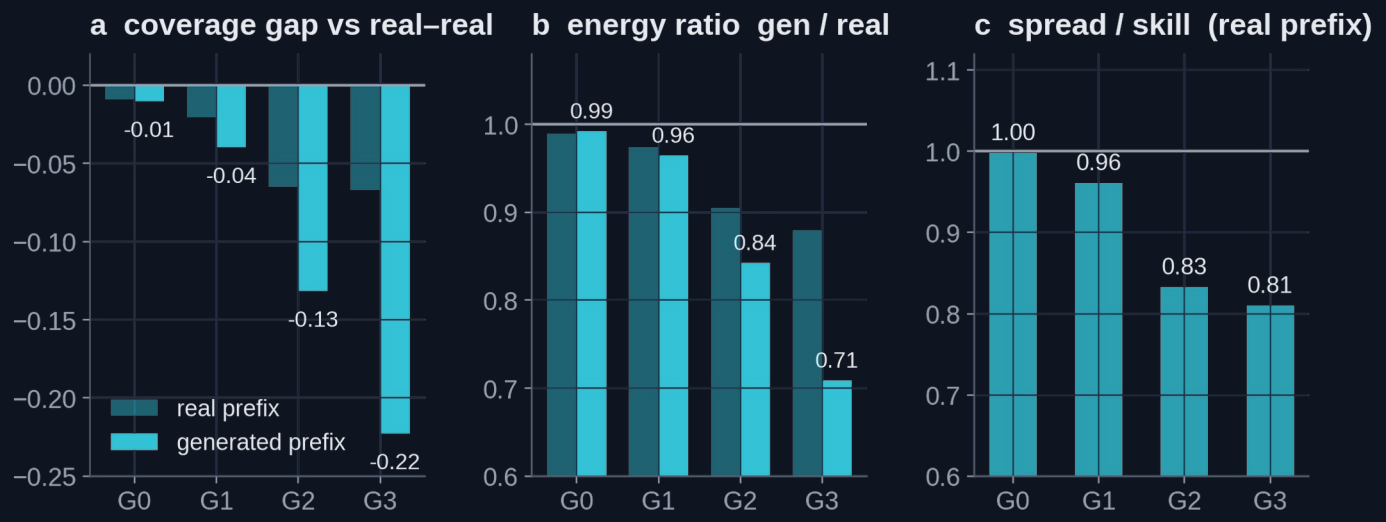


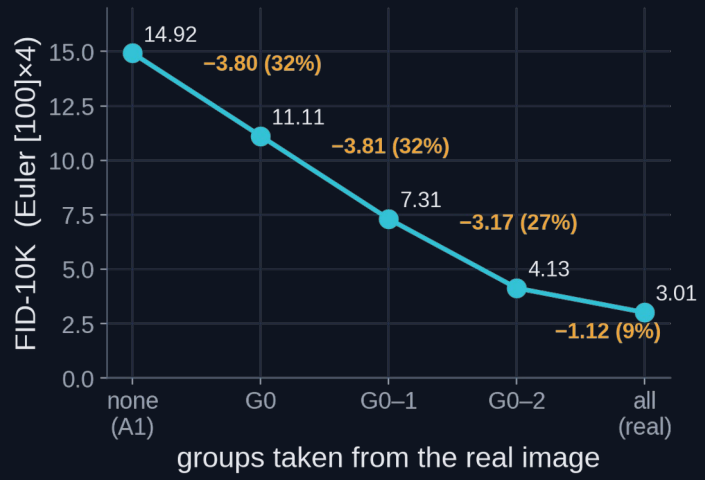
Fairness control: the same solver on R0

- R0 UniPC-15: **FID-50K 8.68**, recall 0.440, 153 GPU-s / 50k, 4.7 ms / image
- A1 stays **1.09× R0 FID**, with **46 vs 15 serial calls**, 1.36× GPU-s, 1.84× latency
- UniPC is not our contribution, and fewer sampling calls do not make training cheaper

FID-50K on paired H100 std seeds (A1 Euler 13.75 here vs 13.69 on A5000 seeds). Latency: isolated H100 NVL, compiled A1 noisy pass + exact prefix cache, R0 eager. *R0 Euler FID from its A5000 run. efficiency/phase2_verdict.json, bench*.json, runs.json

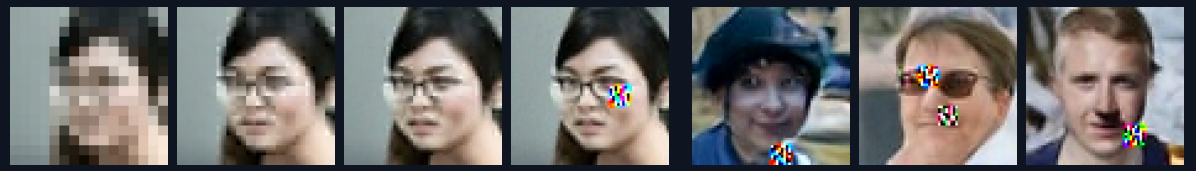
Where A1 fails: later groups lose conditional diversity





real-prefix oracle (Euler sampler)

Tail failure appears exactly when G3 is added (worst of 5k)



G0 ≤G1 ≤G2 all other failing seeds

2.6% of UniPC samples have a G3 coefficient > 5× the data max (113–153× in these); R0 0.5%, A0 0.6%. The trigger is the token's own noise norm (≈3σ), not the prefix.

- **G0 decides layout** (98% of pixel variance) but is adequately modelled: coverage ≈ real–real, spread/skill 0.99
- **Not G0 alone:** the real-prefix oracle splits the gap G0 32 / G1 32 / G2 27 / G3 9% (cumulative, path-dependent — not additive attribution)
- **Later groups are under-dispersed** even with a real prefix (spread/skill .83/.81, energy .90/.88) and it compounds along the chain (energy .84/.71)
- **Inference variance probe:** FID-10K 10.77 → 10.50, recall +0.015 → supports H1 (variance under-estimation)

a–c: efficiency/diag2.json (UniPC [20,10,8,8], std+31M seeds, PCA-64 k=3 recall vs real–real). Oracle: phase1_verdict.json (Euler; real prefixes are also FID-reference images).
Incidence: my recount on 5k paired samples (samples_meta.json).

Only width helped clearly; finer serialization never did

	Lever	Question	Result vs reference	Cost	Verdict
CIFAR	**Width D192 → D256 / D320**	Is A1 capacity-limited?	FID-10K 18.43 → 14.77 / 14.27 @100k	1.27× / 1.62× H100-h	GAIN
	Groups K3 → K7 (same 4×4 DCT)	Do easier transitions help?	22.07 → 21.01 (0.95×)	2.06× H100-h, 700 calls	NEUTRAL
	Full-image DCT, 8 bands	Are global bands better?	51.57 (2.8× A1), P/R .74/.25 — memorises	2.4× H100-h	REGRESSION
	Gaussian G_{-1} context (OT)	Does a shared latent help G0?	19.67 (1.07×)	1.24× H100-h	NEUTRAL
	Context noise σ 0.005	Robustness to own prefix?	EMA 1.02×; raw-weight FID 28.4 → 21.5	≈ 1.0×	NEUTRAL
	A0 refines A1 (no training)	Can joint refinement repair order?	12.59 → 11.64 (~70% of gap); never below A0	+10 calls	PARTIAL
	Depth 13 → 9 blocks	Cheaper calls at equal quality?	raw FID-10K 27.09 → 30.25, recall −.066 @50k	0.74× per step	REGRESSION
	6 radial rings (same 9 coef)	Finer serialization?	raw 27.09 → 38.58 (+42%), +16 calls	1.44× per step	REGRESSION
	8×8 blocks, 36 coef (= 3456)	Fewer, richer tokens?	FID-10K 72 / 83 vs 29 / 27 (EMA / raw); blur	0.43× per step	REGRESSION
FFHQ	**Edit earlier groups (no training)**	Can later bands fix layout?	16.13 → 16.04; random edit of equal norm as good	1.04× sampling	NEGATIVE
	Per-group variance rescale	Late groups too narrow?	FID-10K 10.77 → 10.50, recall +.015	0	SMALL GAIN
	Reverse order G3 → G0	Does the order matter?	FID-10K 46.8 vs 13.7 @100k (each own sampler)	same	ORDER MATTERS
	Per-group data-std precond.	Fix late-group dispersion?	EMA 28.81 → 25.41, raw 26.80 → 26.96; recall +.08 / +.07 @50k	same	BORDERLINE*
	UniPC + per-stage allocation	Was sampling the limit?	FID-50K 13.75 → 9.46, recall +.064	8.0× fewer GPU-s	NOT ARCH.

CIFAR: 100k steps, EMA FID-10K, Euler NFE 100 (scaling/summary.json; refine/results.json). FFHQ screens: 50k, canonical UniPC (efficiency/screens.json); edit: ffhq64/edit; probe & reverse: efficiency/runs.json. Reverse order = 100k screen, stagewise-loss write-up pending. *Candidate A pilot evaluated 11:22 IDT today; not yet adjudicated by the campaign.

Our present frontier: small and close, not cheaper



A Sampling

R0 UniPC-15 has lower FID and lower GPU-s than every A1 point: no A1 sampler is on this frontier.

B Training

At R0's 3.34 H100-h A1 has no UniPC FID-50K; at 5.77 h it is 9.46 vs R0 8.68 at 3.34 h. R0 dominates.

C Size

A1 and R0 are both non-dominated: 4.0× fewer parameters for +0.77 FID and −0.027 recall.

Conclusion we are parameter-efficient and close in FID, but not yet training- or latency-efficient against optimized DCTdiff.

Filled = UniPC (validated per model), hollow = Euler. A0 has FID-10K 15.00 with UniPC-15 but no UniPC FID-50K, so it appears only with Euler. BandJoint: different architecture and hardware budget, shown only in C. phase2_verdict.json, bench*.json, ffhq64/summary.json

Prior art: what we cannot claim to be new

Method	Tokens / representation	What is autoregressed	Within-step spatial model	Stochastic generator
DCTransformer (Nash+ ICML'21)	quantized 8×8 DCT, sparse triples	every non-zero coefficient, low→high	none (one scalar / step)	categorical
JPEG-LM (Han+ 2024)	JPEG file bytes	bytes in codec order	none	categorical (LLM)
DCTdiff (Ning+ ICML'25)	block DCT (we reuse it)	nothing — joint	joint over all tokens	diffusion
GDM (Lee+ 2023)	pixels, frequency groups	groups diffused in sequence	joint	one diffusion process
FAR (Yu+ 2025)	continuous VAE latents, low-pass levels	frequency levels	parallel, per-token head	per-token diffusion MLP
NFIG (Huang+ 2025)	FFT-band-guided residual VQ	frequency / scale stages	parallel per stage (VAR-like)	categorical
SpectralAR (Huang+ ICCV'25)	nested spectral tokens	one spectral token / step	sequential	categorical
MAR · GIVT · JetFormer	continuous latents (+ flow)	spatial tokens	per-token head	diffusion MLP / GMM
Wavelet Flow (Yu+ NeurIPS'20)	Haar wavelet levels	resolution levels	conditional flow per level	normalizing flow
BandJoint (ours, Sep 2026)	JPEG-q90 8×8 DCT, radial bands	5 bands; summed re-encoded prefix	joint per band	per-band diffusion
A1 (ours)	analytic 4×4 DCT, 9 coef, 4 groups	4 coefficient groups	joint 64-position field, causal freq axis	conditional diffusion (46 calls)

Cannot claim

- frequency / coarse-to-fine AR (Nash, FAR, NFIG, SpectralAR, VAR)
- per-stage diffusion over frequency groups (GDM, FAR, BandJoint)
- DCT-space diffusion (DCTdiff); UniPC

Potentially defensible (needs controls)

- causal frequency axis inside one axial net with exact prefix cache
- jointly modelled spatial field per group (vs per-token heads)
- measured per-band evaluation allocation

Missing evidence

- written-up reverse-order mechanism (100k screen only)
- a frontier win vs optimized DCTdiff
- diffusion-free field head (F1) result; 2nd dataset / resolution

Primary sources verified 19 Oct 2026 (arXiv / CVF / PMLR). Closest concept: FAR-Yu (next-frequency + diffusion head); closest to F1: Wavelet Flow, JetFormer, GIVT. Audit: reports/dct_causal/PRIOR_ART_AUDIT.md

Next experiments, and what each outcome would imply

A · Does low→high order matter?

SCREEN DONE

Reverse order, 100k: FID-10K 46.8 vs 13.7 (own samplers), recall .055 vs .43. Next: stagewise-loss write-up; interleaved order if budget.

B · Is diffusion needed per field?

GATES PASSED · QUEUED

F1: axial trunk + conditional affine-coupling flow over each $64 \times C_k$ field (11.66M); 1 trunk + 8 couplings / stage. Strong if ≤ +10% FID at ≥ 2× speed.

C · Coefficient-plane AR (K = 9)

AUTHORIZED · NOT RUN

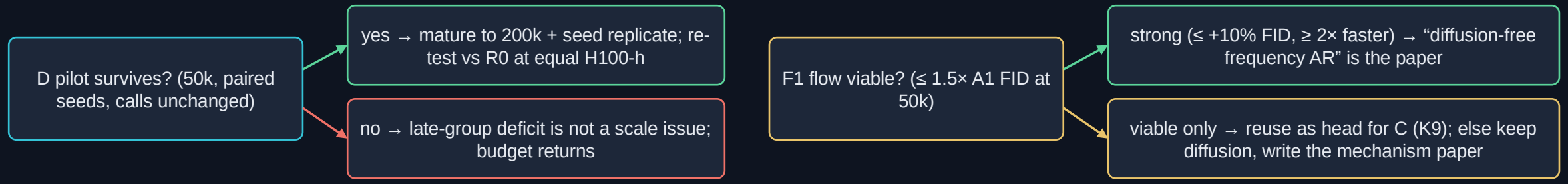
One stage per DCT coordinate, same 3456 values; generator held fixed. Priors are weak: CIFAR K7 neutral, FFHQ K6 rings −42%.

D · Fix late-group dispersion

A EVALUATED · B TRAINING

A (per-group data-std preconditioning, 50k): EMA FID −12%, raw +0.6%, recall +.07–.08 → borderline; verdict pending. B (depthwise 3×3 + 1×1 control) training now.

Decision tree



Order (A) already says the factorization is not arbitrary for this model; whether that makes a contribution depends on B–D beating optimized DCTdiff on some axis other than parameters.

Status at 11:35 IDT 19 Oct from campaign branch 050f9dc + H100: a1_vp done & scored (scr_a1_vp_50000_*, paired std/31M); a1_dw training; f1_flow gate-verified (verify_flow.json), queued; a1_k9 configured. No B or C results.

FOR DISCUSSION

What is the actual paper contribution?

Strongest supported claim

With a matched backbone, explicit low→high frequency causality beats joint generation on FFHQ-64 (A1 15.08 vs A0 18.49 at equal H100-h), and reversing the order collapses it — at 4× fewer parameters than DCTdiff for 1.09× its FID.

Main novelty threat

Next-frequency AR with a diffusion head already exists (FAR-Yu; GDM, NFIG, SpectralAR; Nash for DCT). Our differences — analytic block DCT, joint spatial field, causal axis in one axial net — are design combinations until a controlled mechanism is shown.

Missing decisive evidence

A frontier win over optimized DCTdiff on training or latency; a written-up order mechanism; one more dataset or resolution; a cheaper per-field generator (F1).

Three questions

1. Is explicit frequency autoregression a sufficiently distinct and useful inductive bias — and what would convince a reviewer it is not just a coarse-to-fine schedule?
2. What is the simplest stochastic spatial-field generator that preserves diversity without diffusion's serial cost?
3. Which single experiment would make the architecture both competitive and clearly differentiated from FAR / DCTdiff?

We have promising FFHQ results and a coherent architectural hypothesis, but we have not yet demonstrated a superior overall quality–compute frontier or established a definitive novel mechanism.

Numbers: slides 7, 9, 11, 12. Prior art: slide 13 / appendix A10.

Technical appendix

A1 packing · A2 masks & causality · A3 samplers · A4 evaluation protocol · A5 abandoned designs · A6 throughput & FLOPs · A7 tail artifact · A8 full screen table · A9 paired samples · A10 bibliography

DCT packing: 54 channels, 4 radial groups

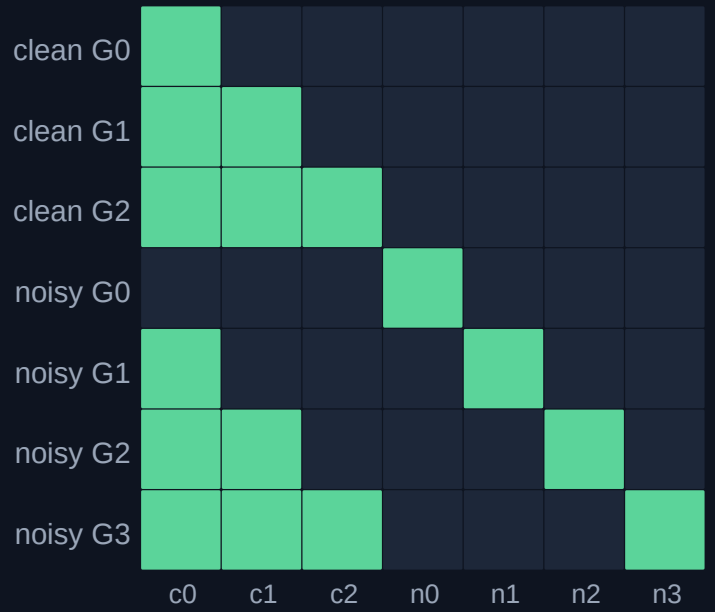
k	(u,v)	group
0	(0,0)	G0
1	(0,1)	G1
2	(1,0)	G1
3	(2,0)	G2
4	(1,1)	G2
5	(0,2)	G2
6	(0,3)	G3
7	(1,2)	G3
8	(2,1)	G3

- Image 64×64 RGB → YCbCr, chroma 4:2:0 → 4×4 block DCT (DCTdiff's own transform, Y_bound η = 480.25, per-coefficient std reweighting, SNR scale 4.0)
- Token = 8×8-pixel area: Y0, Y1, Y2, Y3 (four 4×4 luma blocks) + Cb + Cr (one 4×4 block each) → 6 blocks × 9 coefficients, channel c = block · 9 + k
- Group of coefficient k = u + v of its zig-zag position (u = row / vertical, v = column / horizontal). G0 6 ch, G1 12, G2 18, G3 18 per token
- The 9 retained coefficients include (0,3) but not (3,0): G3 is not orientation-symmetric (zig-zag truncation)
- Coefficients are continuous (mean fractional part 0.247 vs 0.25 uniform; 126,593 of 128,000 values distinct) → no dequantization needed for the flow
- Representation floor: real images through encode/decode give FID-10K 3.01 ("all real" oracle)

DCT_utils.zigzag_order(4)[:9]; libs/axial_freqk.local_dct_stages; PRIOR_ART_AUDIT.md (dequantization check); efficiency/phase1_verdict.json (real_all).

Attention masks and causality checks

A1 training mask (frequency head; rows attend columns)



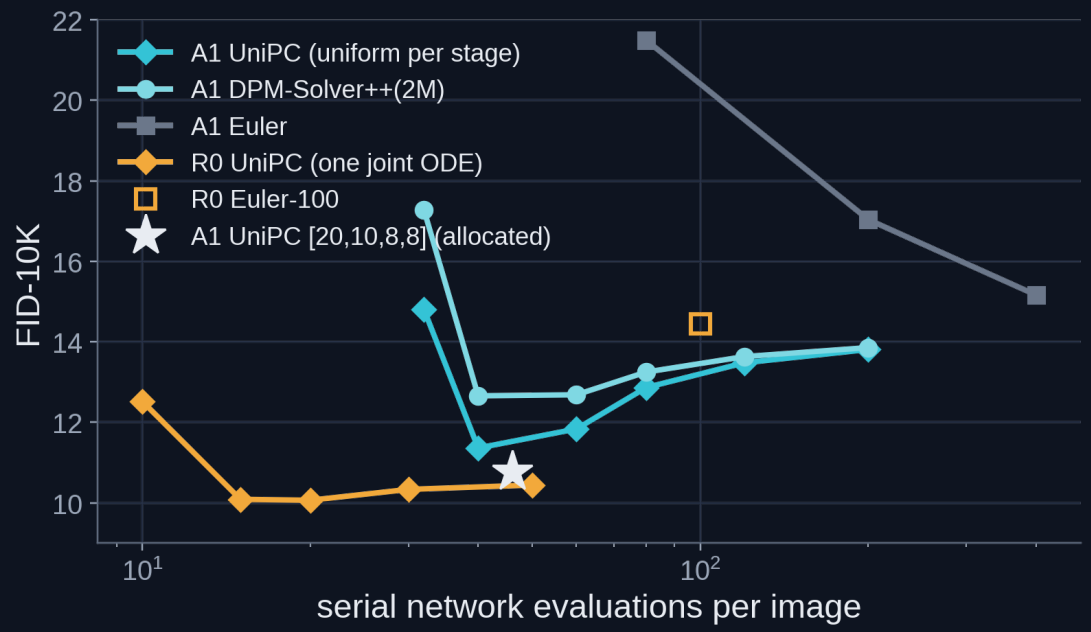
`clean Gₖ ← clean G≤ₖ`
`noisy Gₖ ← clean G<ₖ and itself`

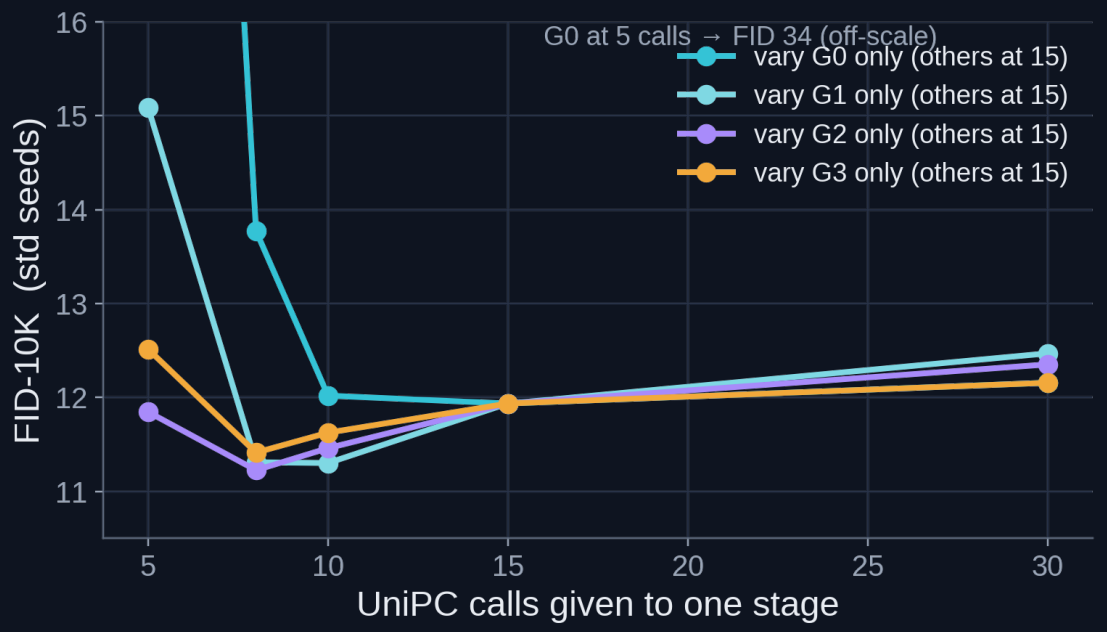
A0: 4 noisy slots, all-true mask. Spatial heads never masked (no spatial order).

- **Jacobian tests (C–G), two attention backends, at init and after 200 AdamW steps:** $\partial\hat{\varepsilon}(G_k)/\partial$ clean $G_k = 0$ exactly; noisy slots never see each other; later stages never reach earlier outputs
- **Exact prefix cache:** clean slots depend only on lower clean slots and t = 0, so their per-layer frequency K/V are computed once per stage; cached vs training-path $\hat{\varepsilon}$ differ ≤ 1e-6
- **Edit probe:** gradients of stage-3 loss into G3's own band are exactly 0
- **Reverse model a1_rev:** leakage test passes; parameter count identical (adapters permuted)
- **BandJoint differs:** it sums the prefix into one token per position and re-runs its encoder per stage; there is no frequency axis inside its transformer

axial/verify_axial_D192.json (B: fmask, inference_vs_training_eps; CDEFG), ffhq64/edit/preflight.json, verify_diffpp.py; report "Implementation audit".

Samplers: equations and evaluation accounting





- **ODE:** VP probability flow of DCTdiff's SDE (SNR scale 4.0); data prediction $\hat{x}_0 = (x_t - \sigma_t \hat{\epsilon})/\alpha_t$; t: 1 → 10^{-3}
- **UniPC-2 bh2** (predictor + UniC corrector reusing the next evaluation), uniform-logSNR grid, final reference Euler step 10^{-3} → 0
- **A1 calls = $\Sigma_k n_k$** (one noisy-slot pass each) + one prefix-cache pass per stage k ≥ 1; GFLOP/image: Euler-400 599, UniPC-46 77; R0 UniPC-15 ≈ 88
- **Allocation curves** vary one stage with the others at 15; joint schedules are not additive ([12,8,6,6] 12.75 vs [20,10,8,8] 10.77)

lsb/dct_causal/ffhq64_eff.py (solvers), efficiency/runs.json (uni_*, fs_*, r0_unipc_*), baseline.json (GFLOP/call 1.47, prefix cache), phase2_verdict.json. FID-10K = mean of std & 31M seed sets where both exist.

Evaluation protocol: q90 reference, FID-10K vs 50K

Primary (FFHQ)	EDM Inception FID vs real_q90_pool: all 70k jpeg-ar q90 FFHQ-64 images (thumbnails128 → LANCZOS 64 → JPEG q90 4:2:0 → decode), exported losslessly; floor FID 1.4e-5
Secondary	FID vs the official EDM FFHQ-64 reference: includes the q90 codec floor (10.77) → R0 35.20, A1-146k 37.27, A1-250k 35.01 · never mixed with primary
Precision / recall	Kynkäänniemi k = 3, Inception pool3, at n = 50k for decisive numbers (at n = 10k real-vs-real recall is only 0.74)
FID-10K vs 50K	FID-10K is biased upward: A1 146k 16.43 (first 10k) vs 15.08 (50k); R0 13.95 vs 12.66 → separate axes, never one plot
Seeds	std = generator 20261017 + shard (paired across samplers); 31M = second independent set. H100 and A5000 CUDA noise streams differ for the same seed
CIFAR-10	DCTdiff's own statistics, which match JPEG-q75 re-encoded images; our R0 250k FID-10K 8.50 vs released checkpoint 8.54 (reproduction OK)
Not comparable	DCTdiff's reported FFHQ-128 6.25 (its q75 stats, pytorch-fid); BandJoint training hours (A5000); any FID-10K against FID-50K

PREREG_FFHQ64.md, PREREG_EFFICIENCY_FFHQ64.md, ffhq64/summary.json, efficiency/baseline.json (note on H100 vs A5000 streams), report "Reproduction and evaluator".

Early architectures we abandoned

Spatial-causal DCTdiff	**WRONG AXIS**	causality over token positions, not frequency; seams at every token boundary at 50k
256-token block-causal staging	**TOO COSTLY**	4 stages × 64 positions as one sequence: correct but 4.18× R0 FLOPs; stopped at 5.7k
Prefix-causal E0 / E1	**OBJECTIVE PROBLEM**	one sampled target group per example; E1 needed 3 calls per solver step
Channel-causal C0 / C1	**CONFOUNDED**	trunk split into 128|256|128 lanes; C1 2.02× worse (recall −0.13), DC output reached 1/16 of weights
Depth-staged P0 / P1	**PAUSED**	groups injected at blocks 6/10/13; frozen at 9.8k when we switched to the axial design
BandJoint (jpeg-ar, Sep 2026)	**SUPERSEDED**	5 radial bands of JPEG-q90 8×8 DCT, summed prefix re-encoded per stage, 122.5M; FID-50K 21.32

Lesson: put frequency on its own axis (slots) instead of serializing tokens or splitting channels — the axial design keeps attention cost at $\text{slots}\cdot(S+1)^2 + (S+1)\cdot\text{slots}^2$ and lets one forward train every conditional.

report.html “Archived 1–4b”; reports/dct_causal/PREREG_CHAN.md, PREREG_DEPTH.md, PREREG_FREQSTAGE.md; jpeg-ar BandJoint campaign memory.

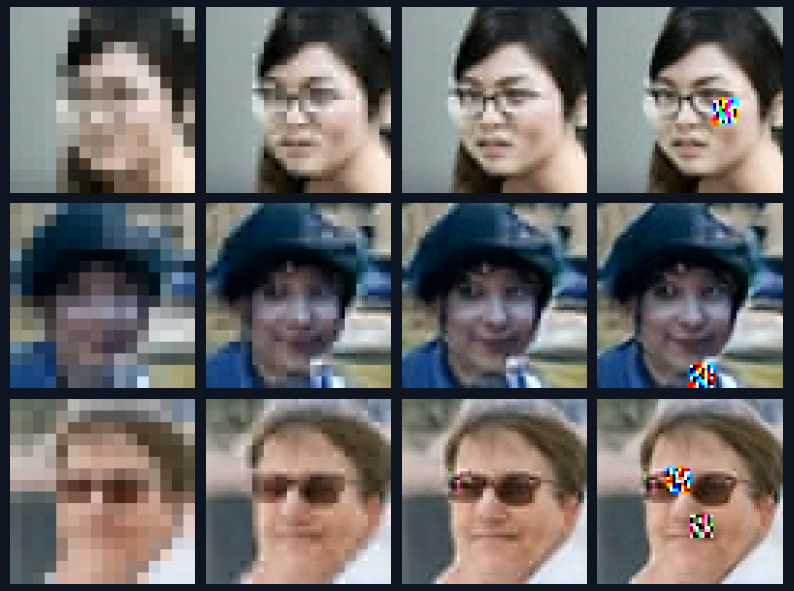
Training throughput and FLOP accounting

	params	H100 steps/s (bs 256)	train GFLOP / image	H100-h to final	sampling GFLOP / image	serial calls
R0 DCTdiff	44.2M	21.05	17.6	3.34 (250k)	≈ 88 (UniPC-15)	15
A0 joint axial	11.09M	14.50	17.8	3.34 (174k)	—	15 (UniPC best)
A1 frequency-AR	11.09M	12.16	31.1	3.36 (146k) · 5.77 (250k)	77 (UniPC-46) · 599 (Euler-400)	46

- A1 trains 4 conditionals per image in one forward over 7 slots × 65 tokens, so 1.75× A0's FLOPs per image; its H100 utilization is higher, which is why A1 wins per hour but only ties per FLOP
- Per call A1 costs 1.47 GFLOP (noisy slot only) vs ≈ 5.9 for R0, but narrow D256 calls are latency/memory-bound: 4× fewer FLOPs per call did not give 4× faster calls
- Evaluation GPU time (A5000 / H100 sampling) is not included in training H100-hours

ffhq64/summary.json (probe_steps_per_s, r0_h100_hours), ffhq64/mature/summary.json (FlopCounterMode, fwd+bwd), efficiency/baseline.json (GFLOP per call / prefix cache), phase2_verdict.json.

The single-token tail artifact



G0 ≤G1 ≤G2 all

- **What:** one token's highest group (G2 on CIFAR, G3 on FFHQ) reaches 2–150× the largest value in 10k real images; all 6 blocks of that token, chroma worst
- **Trigger:** that token's own stage noise norm: 0 failures below 4.5 (of ~255k tokens), 77% at 5–5.25, 100% above 5.25 (CIFAR A1 500k); prefix swaps do nothing
- **Mechanism:** at t = 1 the ε-network under-predicts extreme input noise ($\hat{\varepsilon}/x$ 1.00 up to 2.5σ, 0.91 at 3σ, 0.55 at 5σ); σ/α amplifies the miss; the deterministic ODE locks on
- **Not caused by frequency AR:** CIFAR 100k rates A1 0.85%, A0 0.70%, R0 0.10% — a property of per-stage noise tokens in the axial design
- **Cost:** none measurable in FID (redraw all tokens with norm > 4.5: 15.088 → 15.140). Grows with training. FFHQ UniPC 5k: > 5× data max in 2.6% (A1) vs 0.5% (R0)
- **Fixes (not applied):** truncate per-token noise norm; dynamic $\hat{x}_0$ thresholding; v- or x_0-prediction for noisy slots

narrative G2_DIAG (CIFAR, A1 D192 500k); g2/tail.json, swap_*.json; FFHQ incidence recounted on efficiency j20_10_8_8 / r0_unipc_15 / a0_unipc_15 coefficient arrays (samples_meta.json).

Full screen table (raw numbers)

run	data	steps	params	H100-h	FID-10K EMA	FID-10K raw	recall EMA / raw	sampler
A1 D192	CIFAR	100k / 250k / 500k	6.24M	1.42 / 3.64 / 7.34	18.43 / 15.19 / 15.09	28.44 (100k)	.494 (100k)	Euler 100/stage
A1 D256	CIFAR	100k / 185k / 250k	11.08M	1.80 / 3.32 / 4.48	14.77 / 12.81 / 13.16	18.18 / 19.70 / 21.40	.537 / .564 / .560	Euler 100/stage
A0 D256	CIFAR	100k / 200k / 250k	11.08M	1.65 / 3.30 / 4.12	13.99 / 11.49 / 10.97	17.04 / 16.81 / 17.30	.557 / .575 / .589	Euler 100
R0 (ours)	CIFAR	100k / 250k	44.2M	1.34 / 3.35	12.67 / 8.50	16.05 / 18.31	.565 / .600	Euler 100
A1 D320	CIFAR	100k	17.29M	2.30	14.27	21.52	.550	Euler
4×4 K3 / K7	CIFAR	100k	6.24M	1.40 / 2.88	22.07 / 21.01	25.88 / 25.31	.472 / .473	Euler
full-image K8	CIFAR	100k	6.31M	3.43	51.57	59.12	.246	Euler
G_{-1} / σ 0.005	CIFAR	100k	6.24M	1.76 / 1.48	19.67 / 18.75	30.61 / 21.48	.500 / .497	Euler
A1 D256 (ctrl)	FFHQ	50k	11.09M	≈1.15	28.84	27.09	.204 / .315	UniPC 46
L8 · K6 · 8×8 k36	FFHQ	50k	7.67 / — / 11.14M	0.85 / — / 0.52	48.43 / 52.08 / 72.41	30.25 / 38.58 / 82.67	.053 / .053 / .011 (EMA)	UniPC
A1 vs reverse	FFHQ	100k	11.09M	≈2.3	13.69 vs 46.77	22.94 vs 69.87	.428 vs .055	own best
A0 D256	FFHQ	174k	11.09M	3.34	15.00 (UniPC-15)	—	.393	UniPC 15

scaling/summary.json (arms, d256_maturity, a1), efficiency/screens.json, efficiency/runs.json (rev100_*, a1100_*, a0_unipc_*). FFHQ 100k reverse comparison: FID-10K mean of std/31M where both exist; reverse std only.

Same initial noise, different samplers and models

A1 250k · Euler 400

A1 250k · UniPC 46

R0 250k · UniPC-15

A0 174k · UniPC-15

A1 100k · UniPC 46

reverse 100k · UniPC



Std seeds 0–11, H100, uncurated. Euler → UniPC keeps identity and sharpens. The two joint models (R0, A0) map the same noise to nearly the same face; A1 diverges more, consistent with its layout being set by the G0 noise slice alone. Reverse order keeps faces but loses colour and diversity.

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